

NutriFlow AI — Comprehensive Engineering & User Manual

Cloud-Native Precision Health Assessment, AI Nutrition Engine & Calorie-Targeted Food Delivery Platform

Version: 2.0.0 Cloud-Native	Backend: FastAPI / Python 3.10+	Database: SQLite / PostgreSQL	Frontend: Vanilla JS SPA + Leaflet
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1. Executive Summary & Why NutriFlow AI is Unique

Modern fitness and health management faces a fundamental industry disconnect known as the **'Execution Gap'**: traditional calorie tracking applications (such as MyFitnessPal or HealthifyMe) only record nutritional intake *after* consumption, while commercial food delivery applications (like Zomato, Swiggy, or UberEats) prioritize consumer cravings without portion control or macro calibration. Users are left with the tedious cognitive burden of manually computing macros, searching menus, estimating portion sizes, and adjusting ingredients.

NutriFlow AI completely solves this paradigm by uniting clinical diagnostic calculation, automated chef meal planning, and on-demand delivery into an integrated closed-loop platform:

- Precision Clinical Engine:** Computes exact physiological parameters (Mifflin-St Jeor BMR, Physical Activity TDEE, Clinical BMI, and goal deficits/surpluses) rather than arbitrary caloric guesses.
- Deterministic Calorie-Targeted Meal Formulation:** Automatically distributes daily calories across 4 distinct meal budgets (Breakfast 25%, Lunch 35%, Dinner 30%, Snack 10%) and matches them with chef-curated whole foods.
- Direct Commercial Fulfillment in Indian Rupees (₹ INR):** Allows users to order a complete 4-meal daily regimen or single dishes in 1-click with real-time bill calculations (subtotal, 5% GST, ₹0 delivery) and address management.
- Live Interactive GPS & Route Tracking:** Features an embedded Leaflet/OpenStreetMap engine that captures real live GPS or 1-click Delhi NCR presets and renders animated rider transit stages.
- Conversational Gemini AI Nutrition Coach:** Integrated multi-turn Google Gemini AI for customized recipe substitutions, macro explanations, and lifestyle Q&A;

2. Recommendation Engine & Mathematical Architecture

NutriFlow AI does not rely on random guessing. Every recommendation is derived from validated physiological science through a multi-step deterministic pipeline:

Metric / Stage	Governing Formula / Clinical Logic	Application in NutriFlow AI
Basal Metabolic Rate (BMR)	Mifflin-St Jeor Equation: - Men: $BMR = 10W + 6.25H - 5A + 5$ - Women: $BMR = 10W + 6.25H - 5A - 161$ (W in kg, H in cm, A in years)	Calculates baseline calories burned at complete metabolic rest with 95% clinical accuracy.
Total Daily Energy Expenditure (TDEE)	TDEE = BMR × Activity Multiplier: - Sedentary: 1.20 Light: 1.375 - Moderate: 1.55 Active: 1.725 - Very Active: 1.90	Calculates real-world energy expenditure factoring occupational and exercise demands.

Caloric Target & Goal Offsets	Target = TDEE + Goal Delta: • <i>Weight Loss</i>: -500 kcal/day (~0.5 kg fat loss/week) • <i>Muscle Hypertrophy</i>: +350 kcal/day (lean mass gain) • <i>Maintenance</i>: 0 kcal offset	Prevents starvation responses while driving sustainable body composition results.
Macronutrient Distribution	Macro Split Target: • <i>Protein</i>: 25% of kcal (4 kcal/g) • <i>Carbohydrates</i>: 50% of kcal (4 kcal/g) • <i>Healthy Fats</i>: 25% of kcal (9 kcal/g)	Calculates daily gram targets for muscle synthesis, sustained glycogen, and hormonal health.
Meal Calorie Budgets	Portion Allocation: • <i>Breakfast</i>: 25% of daily calories • <i>Lunch</i>: 35% of daily calories • <i>Dinner</i>: 30% of daily calories • <i>Snack</i>: 10% of daily calories	Partitions total daily target into 4 distinct portion budgets for automated dish matching.
Multi-Offset Dish Rotation	Permutation Algorithm: • <i>Breakfast Idx</i>: variation % N • <i>Lunch Idx</i>: (variation × 2 + 1) % N • <i>Dinner Idx</i>: (variation × 3 + 2) % N • <i>Snack Idx</i>: (variation × 4 + 3) % N	Cycles through 60+ database dishes with non-overlapping step offsets on every 'Regenerate' click.

3. Interactive UI Controls & Button Functionality Guide

The NutriFlow AI single-page dashboard provides rich interactive controls designed for zero-friction user workflows. Below is a comprehensive breakdown of all buttons and their underlying technical actions:

UI Button / Element	Location	Functionality & Underlying Technical Action
'Update Metrics'	Header Navigation	Opens the modal dialog allowing users to modify Age, Gender, Height, Weight, Activity Level, Goal, and Diet. On submit, re-triggers NutritionEngine.evaluate_health() and updates all metrics & meal cards.
'Regenerate Meals'	Meal Plan Section Header	Increments currentMealVariation, triggers a rotating spin animation, and queries the backend with the new variation offset. Pulls fresh dishes from the 60+ database catalog and triggers a floating success toast.
'Order Full Day Plan'	Meal Plan Section Header	Simultaneously adds all 4 meals (Breakfast, Lunch, Dinner, Snack) into the cart in a single click. Categorizes each item with badges, updates total calories/pricing in ₹ INR, and smoothly slides open the cart drawer.
'+ Add Meal'	Each Individual Meal Card	Adds the specific selected meal to the cart, increments the navigation cart counter badge, recalculates totals, and opens the drawer with instant feedback.
'+' / '-' Qty Buttons	Delivery Cart Drawer	Dynamically increments or decrements dish quantity. Recalculates item subtotals, overall calorie count, 5% GST, and grand total. Automatically removes item when quantity reaches 0.
'Clear All'	Cart Drawer Header	Empties the entire delivery cart in one click, resets badge counters to zero, and displays the empty state placeholder.
'Place Order & Track Live'	Cart Drawer Footer	Validates address, creates an order via POST /api/v1/orders/, generates a unique Order ID, opens the Big Live Map modal, and initiates simulated rider GPS dispatch.
'Live Map Tracking' / Map Window	Top Nav & Dashboard Card	Opens the full-screen interactive Leaflet map modal with auto-tile rendering, live GPS geolocation button, quick Delhi-NCR address chips, delivery time estimation, and restaurant routing.

'Ask Coach' (AI Nutritionist)	AI Coach Floating/Card Widget	Sends multi-turn conversational health queries to the Google Gemini API (gemini-1.5-flash / gemini-2.0-flash) passing the user's current health metrics as system context.
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4. Nutritious Food Catalog & Dietary Diversity (60+ Meals)

The platform includes an extensive, curated database of 60 high-protein, calorie-calibrated meals split evenly across 4 categories and all dietary preferences (Vegetarian, Vegan, Non-Vegetarian, Keto):

Meal Category	Caloric Budget	Sample Chef Dishes (High Protein & INR Pricing)	Supported Diets
Breakfast (15)	320 – 510 kcal	Moong Dal & Paneer Chilla (₹179), Avocado Poached Egg Sourdough (₹249), Oats & Whey (₹189), Steamed Oats Idli (₹169), Masala Omelette (₹199), Smoked Chicken Wrap (₹279), Nutty Banana Oatmeal (₹189).	Veg, Vegan, Non-Veg, Keto
Lunch (15)	460 – 620 kcal	Palak Paneer with Multigrain Rotis (₹279), Soya Chunk Curry & Brown Rice (₹229), Amritsari Rajma Quinoa (₹239), Tandoori Chicken Thali (₹349), Salmon Quinoa Bowl (₹449), Lemon Herb Chicken (₹329).	Veg, Vegan, Non-Veg, Keto
Dinner (15)	390 – 580 kcal	Yellow Moong Dal Khichdi (₹219), Lentil Dal Makhani (₹239), Paneer Tikka Masala (₹289), Smoked Chicken Tikka (₹329), Mutton Seekh Platter (₹399), Basa Fish & Veggies (₹369), Baingan Bharta (₹219).	Veg, Vegan, Non-Veg, Keto
Snacks (15)	140 – 230 kcal	Peri-Peri Roasted Makhana (₹109), Sprouted Moong Chaat (₹129), Masala Boiled Eggs (₹99), Artisanal Energy Bites (₹119), Whey Protein Smoothie (₹179), Greek Yogurt & Pistachios (₹159), Edamame (₹149).	Veg, Vegan, Non-Veg, Keto

5. Prerequisites & Installation Guide

Follow the steps below to run the complete NutriFlow AI platform locally on Linux, macOS, or Windows:

A. System Prerequisites:

- **Python:** Python 3.10, 3.11, or 3.12 installed.
- **Package Manager:** pip or ultra-fast uv installer.
- **Git:** Version control CLI.
- **Web Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge.
- **(Optional) Gemini API Key:** Set GEMINI_API_KEY environment variable for AI Coach.

B. Step-by-Step Terminal Commands:

```
# 1. Clone or navigate to the project directory
cd /path/to/project

# 2. Create a virtual environment (using standard venv or uv)
python3 -m venv .venv
source .venv/bin/activate # On Windows: .venv\Scripts\activate

# 3. Install backend dependencies
pip install fastapi uvicorn sqlalchemy pydantic pydantic-settings python-multipart python-jose bcrypt
httpx jinja2

# 4. (Optional) Set your Gemini API Key for AI Nutrition Coach
export GEMINI_API_KEY="your-google-gemini-api-key"

# 5. Start the FastAPI ASGI server with auto-reload
uvicorn app.main:app --app-dir backend --reload --host 127.0.0.1 --port 8000

# 6. Open your web browser and navigate to:
# Application Dashboard: http://127.0.0.1:8000/
# Interactive Swagger API Docs: http://127.0.0.1:8000/docs
```

6. Step-by-Step GitHub Push Guide

To publish your codebase to GitHub, follow these exact terminal commands:

```
# Step 1: Open terminal in project root directory
cd /path/to/project

# Step 2: Initialize git repository on 'main' branch
git init -b main

# Step 3: Stage all project files (safe .gitignore prevents large venvs from being added)
git add .

# Step 4: Commit all full-stack project files
git commit -m "feat: NutriFlow AI - Cloud-Native Precision Health, AI Nutrition & Delivery Platform"

# Step 5: Create a new repository on GitHub (e.g. https://github.com/new)
# Then link your local repo to GitHub (replace and )
git remote add origin https://github.com//.git

# Step 6: Push your code to GitHub
git push -u origin main
```

7. Cloud-Native DevOps & Deployment Architecture

NutriFlow AI is enterprise-ready and engineered for containerized multi-cloud deployment:

- **Docker Containerization:** Multi-stage lightweight Python Dockerfile with non-root security contexts.
- **Kubernetes (EKS / GKE):** Production manifests in /devops/k8s including Deployments, ClusterIP Services, NGINX Ingress, Horizontal Pod Autoscalers (HPA), and ConfigMaps.
- **Terraform Infrastructure as Code (IaC):** Modular AWS infrastructure in /devops/terraform provisioning VPCs, Subnets, Security Groups, and EKS managed node groups.
- **Ansible Automation:** Playbooks in /devops/ansible for server hardening, Docker runtime setup, and automated CI runner provisioning.

- **GitHub Actions CI/CD:** Pipeline in .github/workflows running pytest suites, linting, Docker build/push, and zero-downtime Kubernetes rolling deployment.